

UNIVERSITY POMPEU FABRA

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Analyzing and Predicting University Rankings (QS and THE): Building a Strategy for UPF

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Abstract

Global university rankings have become one of the most visible reference points in higher education. Students, employers, governments and universities themselves treat them as signals of institutional quality, which creates a self-fulfilling dynamic: a higher rank attracts better students, more funding and greater prestige, reinforcing the conditions that produced the rank in the first place. Yet rankings are not neutral measurements. They are constructed outputs of opaque methodological choices, produced by organisations that also sell advisory services to the universities they rank.

This thesis treats Times Higher Education (THE) and QS as systems of strategic incentives rather than measures of quality, and asks what each ranking actually rewards, what are the structural differences among them, and how easily institutions (in particular, UPF) can move through it. The analysis moves from diagnosis to strategy in three steps. It first examines the published pillars directly, using descriptive analysis, principal components and clustering, to identify what each ranking rewards. It then tests how solid the resulting positions are, through Monte Carlo weight simulations, a score-gap mobility analysis of the full THE distribution, and a dynamic panel of THE between 2017 and 2024. Finally, both rankings are modelled using proxy variables that universities can actually act on. Universitat Pompeu Fabra runs through the whole analysis as a case study, and the findings are translated into a concrete strategy for its position in each ranking.

The results show that QS and THE do not rank the same model of excellence: THE behaves as a single underlying type of excellence, while QS rewards a broader portfolio of reputation, employability, internationalisation and research visibility. QS positions are not equally precise across the distribution, this is: the global elite is highly stable, but mid-ranked universities can shift by over a hundred places under plausible weight changes. THE is not an evenly spaced ladder either: elite ranks are separated by wide score gaps, while lower ranks are compressed and more fragile. Positions are also strongly persistent over time, so measurable improvement registers slowly and only along specific dimensions.

For UPF, this produces two different strategic diagnoses of the same institution: a university that is research-competitive in both systems, but constrained in QS by weak external signalling (reputation and employer recognition) and in THE by limited institutional depth (teaching capacity and research environment relative to its research impact).

The project is concluded with the delivery of a practical tool designed to give continuity and operational value to the analysis beyond the scope of this thesis. Rather than treating the results as a static diagnosis, the tool allows members of UPF to explore additional scenarios, modify ranking assumptions, compare alternative views, and assess how changes in specific indicators could affect the university's position. This makes the analysis reusable for future institutional questions and potential new strategic interests. The tool's main functionalities, structure and usage are described in detail in the accompanying manual, delivered in parallel with this thesis. The tool is available at: <https://upf-rankings.vercel.app/>

“Virtute enim ipsa non tam multi praediti esse quam videri volunt.”

“Few wish to be endowed with virtue rather than to seem so.”

— Cicero, *Laelius de Amicitia* (ch. 98)

1 Introduction

University rankings have become one of the most visible instruments through which higher education institutions are compared. Systems such as the QS World University Rankings and the Times Higher Education World University Rankings publish annual lists that order thousands of universities along a single ordinal scale. What began in the early 2000s as a relatively narrow consumer tool, designed primarily to inform prospective students about institutional reputation, has since expanded into something much broader. Today, rankings shape how universities present themselves externally, how institutional leaders benchmark performance, how governments interpret national competitiveness, and how partners, funders and employers assess academic prestige. As Hazelkorn (2008) argues, league tables and ranking systems have moved well beyond consumer information and now influence institutional behaviour, decision-making, stakeholder perceptions and higher education policy.

Rankings matter so much partly because of what a good position brings with it. A university that ranks well attracts better students, more funding and more willing partners, and all of that makes it easier to rank well again the following year. The opposite is also true: when a university slips, fewer students apply, it loses visibility, and it negotiates from a weaker position with partners and funders. A ranking is therefore never just a photograph of where a university stands at a given moment. It also changes how others treat that university, and that treatment ends up shaping where it stands next.

This creates a paradox that the academic literature has documented in detail. The European University Association’s RISP project, based on responses from 171 higher education institutions across 39 countries, found that 86% of respondents monitored their ranking position, 60% reported that rankings played a role in institutional strategy, and 82% advertised their ranking position in some form. Institutions also identified prospective students, researchers, partner universities, ministries, funding bodies, employers and sponsors as stakeholders influenced by ranking results (Hazelkorn et al., 2014). Universities frequently criticise rankings for methodological simplification, yet they continue to monitor, communicate and respond to them because the reputational and strategic costs of ignoring them are too high. Universities of the Netherlands (2023) makes the same point from a different angle: one-dimensional league tables claim to summarise the overall performance of a university in a single score, but universities perform several distinct core tasks (education, research, social impact, etc.) and there is no universally accepted way to combine these dimensions into one measure of “quality”. Any weighting system necessarily favours some institutional profiles over others. Rankings, in this sense, do not simply measure quality. They define a particular version of it.

The problem is deepened by two structural features of the ranking industry. The first is opacity. Ranking providers publish broad pillar weights and general methodological notes, but the precise aggregation rules, the score transformations applied to raw indicators, and the marginal effect of any single dimension on the final position are not transparently disclosed for most indicators listed. Universities know, in approximate terms, what is being measured, but not how much each actionable input/ratio actually moves their position. The second is a structural conflict of interest. The same organisations that produce the rankings also sell paid advisory and consultancy services to the universities they rank, offering tailored guidance on how to improve positioning. This combination (opaque methodology, produced by organisations with a commercial incentive to keep that methodology partly opaque) means that the evaluation system shaping institutional behaviour and resource allocation is not fully accountable to the

institutions it affects.

There is a further complication: published scores are not even stable over time. In THE, for instance, new universities enter the ranking every year, most of them near the bottom, and the 0–100 scale is recalculated in each edition. The result is a quiet score inflation: the average score rises year after year, so a university can see its score improve while gaining no ground on anyone. Reading a ranking trajectory therefore means separating real progress from this background drift, and the historical part of this thesis does exactly that.

This thesis starts from these observations. If rankings influence institutional behaviour through opaque methodologies that universities cannot fully reverse-engineer on their own, then the methodology is not a technical detail. It is the core of the incentive structure facing universities. Rankings define what counts as performance, assign value to different dimensions of that performance, and translate those dimensions into positional advantage. A university may be rewarded or constrained not only because of how well it performs, but because of how closely its profile matches the model of quality embedded in a particular ranking.

The central research question of this thesis is therefore: do global university rankings measure a common concept of university quality, or do they construct different models of quality that create unequal incentives, robustness and mobility conditions for universities?

This question has a direct applied counterpart, which gives the thesis its second objective: if each ranking constructs its own model of quality, what does each one reward and penalise in the specific case of Universitat Pompeu Fabra, and which strategic levers are most likely to translate into real ranking gains for it?

To answer this question, the thesis analyses QS and Times Higher Education using multivariate analysis, simulations and mobility measures. Instead of treating a university's rank as a number to be explained, it treats each ranking as a system of incentives and reverse-engineers its logic from the data it publishes: what it actually rewards, how stable its positions are when the methodology changes, and how much improvement it takes to move up once a methodology is fixed. As a final step, both rankings are modelled with proxy variables that universities can act on, which is what allows the thesis to end with a concrete strategy for UPF in each system.

Universitat Pompeu Fabra is used as a running case study throughout the analysis. This is not only an illustrative choice but a motivating one. UPF is a young, research-intensive university that performs strongly along some dimensions (particularly research output and international faculty) while remaining more constrained in others. Its position therefore makes it a useful institution through which to test whether QS and THE produce the same strategic diagnosis of the same university, and to identify which strategic levers are most likely to translate into ranking gains given its specific profile.

The empirical analysis proceeds in four steps. First, principal component analysis and clustering are used to identify different institutional profiles and to study the QS and THE as revealed strategic models, showing whether the two rankings reward the same underlying concept of quality. Second, a Monte Carlo sensitivity analysis of QS tests how robust university positions are when indicator weights are perturbed. Third, a score-gap mobility analysis of THE 2024 calculates how many overall score points universities need to gain to move up 10, 25 or 50 positions, revealing whether ranking positions are evenly spaced or whether some parts of the distribution are structurally more mobile than others. Fourth, a dynamic panel model of THE between 2017 and 2024 examines how persistent positions are over time and how quickly

real improvements, once system-wide score inflation is removed, translate into observable rank movement.

2 Ranking methodologies: THE and QS

Before analysing university performance across the main international rankings, it is essential to understand the official methodologies used by QS and Times Higher Education World University Rankings.

Both rankings are based on an overall score for each institution, which determines its final position in the ranking. This overall score is calculated as a weighted aggregation of several performance indicators, named pillars. In the case of THE, each pillar is further composed of several sub-pillars, which are the actual indicators used in the score calculation; the full breakdown of sub-pillar weights and descriptions is provided in Appendix A.1. QS, by contrast, does not publish a sub-pillar decomposition, its pillars are the finest level of disaggregation made available.

THE

Table 1: THE World University Rankings: pillars, weights and sub-pillars.

Pillar	Weight	Description / Sub-pillars
Teaching	29.5%	Measures the quality of the university’s learning environment and its teaching resources. <i>Sub-pillars:</i> Teaching reputation (15%), Student-to-staff ratio (4.5%), Doctorate–bachelor ratio (2%), Doctorate–staff ratio (5.5%), Institutional income (2.5%).
Research environment	29%	Measures the university’s research capacity and resources. <i>Sub-pillars:</i> Research reputation (18%), Research income (5.5%), Research productivity (5.5%).
Research quality	30%	Measures the quality and impact of the research produced. <i>Sub-pillars:</i> Citation impact (15%), Research strength (5%), Research excellence (5%), Research influence (5%).
International Outlook	7.5%	Measures how international the university is in terms of students, staff and research collaboration. <i>Sub-pillars:</i> International students (2.5%), International staff (2.5%), International co-authorship (2.5%), Students abroad (0%).
Industry	4%	Measures the university’s connection with industry through knowledge transfer, industry income and patents. <i>Sub-pillars:</i> Industry income (2%), Patents (2%).

Table 2: QS World University Rankings: pillars and weights.

Pillar	Weight	Description
Academic reputation	30%	Measures global academic prestige based on academics' nominations of excellent universities.
Employer reputation	15%	Measures employers' perception of which universities produce strong graduates.
Faculty-student ratio	10%	Proxy for teaching capacity; faculty resources relative to student numbers.
Citations per faculty	20%	Measures research impact relative to academic staff size.
International faculty ratio	5%	Share of foreign-national faculty among total faculty.
International student ratio	5%	Share of foreign-national students among total students.
International research network	5%	Measures diversity of sustained international research partnerships.
Employment outcomes	5%	Measures graduates' employability and alumni impact.
Sustainability	5%	Measures sustainability performance of the institution.
Institutional students diversity	0%	Measures the ratio of international students to overall students as well as the diversity of nationalities that those students are from.

At first glance, QS appears to cover more characteristics than THE, as it uses ten indicators compared to five broader pillars. However, this difference is mainly structural: several QS indicators are reported separately, while THE groups similar dimensions into wider categories. The main contrast is that QS gives a more explicit role to reputation, especially through academic and employer surveys, whereas THE places greater emphasis on teaching and research-related performance. Therefore, QS can be seen as more reputation- and employability-oriented, while THE provides a broader academic and research assessment. It is also worth noting that the “International Student Diversity” indicator currently has a 0% weight and does not affect the overall QS score. Also is notable that THE does not account for any kind of sustainability as QS does.

Having outlined how each ranking constructs its overall score, it is useful to first situate UPF within this landscape. The figures below show UPF's trajectory in THE and QS over time, together with its current position relative to other Catalan universities (UB, UAB and UPC). These figures provide a first descriptive overview before turning to the data and methodology used in the rest of the thesis.

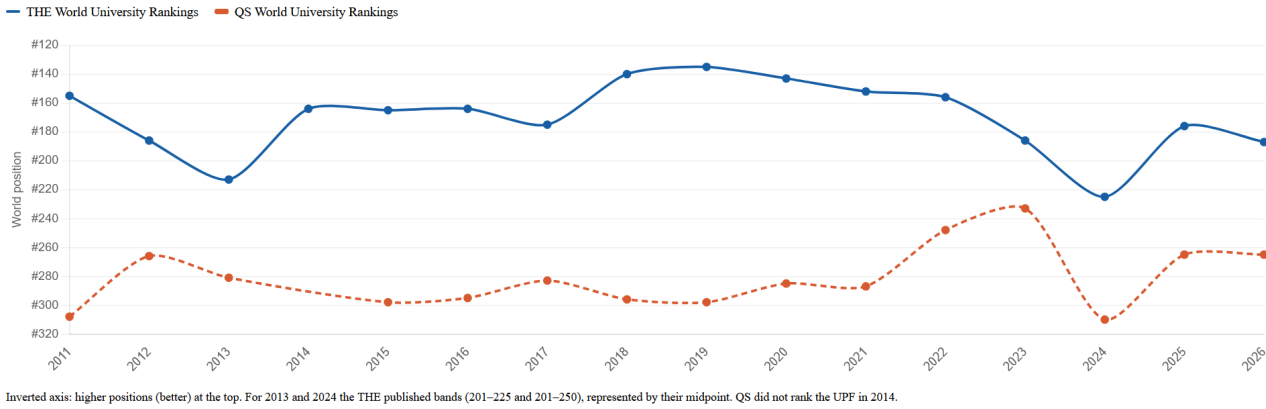


Figure 1: UPF's position over time in THE and QS. Inverted axis: higher positions (better) at the top. For 2013 and 2024 the THE published bands (201–225 and 201–250) are represented by their midpoint. QS did not rank UPF in 2014.

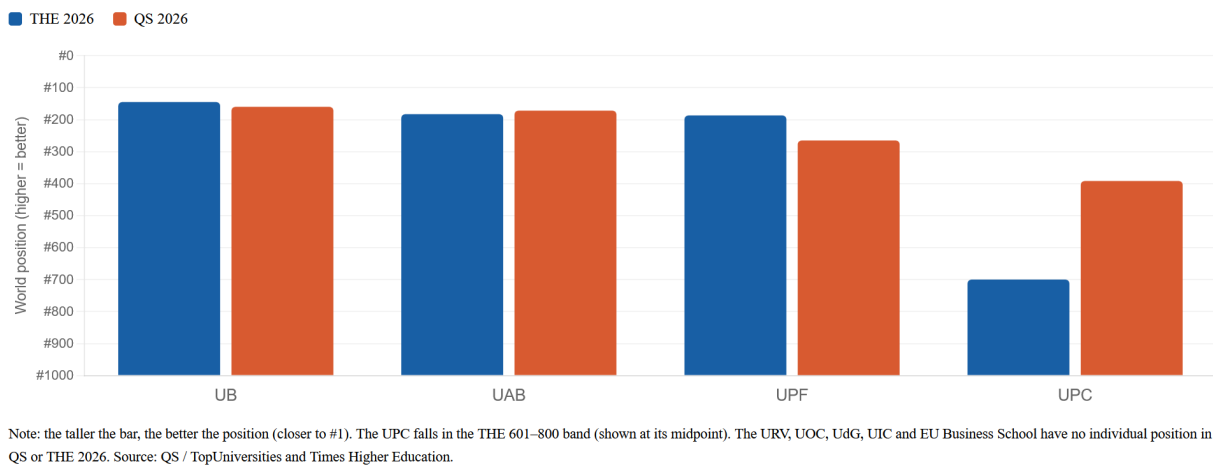


Figure 2: Current position of UPF relative to other Catalan universities (THE 2026 and QS 2026). The taller the bar, the better the position (closer to #1). The UPC falls in the THE 601–800 band (shown at its midpoint). The URV, UOC, UdG, UIC and EU Business School have no individual position in QS or THE 2026. Source: QS / TopUniversities and Times Higher Education.

Overall, the figures suggest that, in both THE and QS, UPF ranks behind UB and UAB, making it the third-best positioned Catalan university among the institutions compared. This gap is especially visible in QS, where UPF is further from UB and UAB than in THE.

However, this general picture changes when rankings are analysed by subject. According to the QS World University Rankings by Subject 2025, UPF performs particularly well in Social Sciences and Management, where it is ranked 84th worldwide. In this field, it is the best-positioned Catalan university, ahead of UB (117th) and UAB (129th). By contrast, in Natural Sciences the situation is reversed: UB ranks 60th globally, UAB 133rd and UPF 243rd.

3 Data Used

The dataset used in this thesis combines two types of information: (i) the official data published by the QS and Times Higher Education (THE) rankings, and (ii) a set of supplementary descriptive variables collected independently from external sources.

The ranking providers publish each university’s position or rank band, overall score where available, and pillar-level scores, with scores generally reported on a 0–100 scale. Since these scores are highly aggregated and their underlying composition is not always fully observable, additional descriptive variables were collected to better understand the factors associated with each pillar. This second dataset, composed of descriptive or proxy variables, will later be used to build the final predictive model of the ranking. Finally, additional demographic and geographic variables were also included.

3.1 Ranking Variables

For each institution, the dataset contains the pillars described in the previous section and the final score (the aggregation of the pillars).

This data was obtained from the official QS and THE ranking websites, through a combination of web scraping and manual collection of published rankings. Since neither provider distributes a consolidated database, the procedure collected all available indicator scores for each university across editions. In addition, the support of UPF’s Strategy, Data and Quality Service (SEDQ), specifically its Data and Rankings team, was relied upon to obtain certain data.

Three datasets are used throughout the thesis:

- **QS 2024**, used in the principal component analysis, clustering analysis, and Monte Carlo simulations. The original dataset contains 1,168 universities. After removing observations with missing values, the final sample contains 1,085 institutions.
- **THE 2024**, used in the cluster comparison, mobility analysis, and predictive modelling. The dataset contains 1,904 universities.
- **THE Panel 2016–2024**, used in the historical validation exercises and the dynamic panel model. The panel contains 9,332 university-year observations across 1,683 institutions and is unbalanced because universities do not appear in every edition.
- **QS/THE 2024–2026**, included in the practical tool as default scenarios for the what-if and variance analyses.

3.2 Supplementary Descriptive (Proxy) Variables

The following table summarizes the main supplementary descriptive variables used as proxies for the ranking pillars. For each variable, the table indicates the pillar or dimension it is intended to approximate and its data source.

Table 3: Supplementary descriptive variables used as pillar proxies.

Variable	Main proxy for	Source
Student-to-Staff Ratio	Teaching / Learning environment	THE official dataset
Estimated Staff	Teaching / Institutional capacity	THE official dataset
International Students (%)	International Outlook	THE official dataset
Female Students (%)	Institutional diversity	THE official dataset
Total Citations	Research Quality / Research impact	OpenAlex/ROR public databases
Total Publications	Research Environment / Research output	OpenAlex/ROR public databases
H-Index	Research Quality / Research impact	OpenAlex/ROR public databases
5-Year Publication Growth (%)	Research Environment / Research growth	OpenAlex/ROR public databases
5-Year Citation Growth (%)	Research Environment / Research growth	OpenAlex/ROR public databases
Research Productivity	Research Environment / Research productivity	Calculated: publications per staff
International Co-authorship Papers	International Outlook	OpenAlex/ROR public databases
Alumni Impact Index (0-1000)	Employment Outcomes	QS web scraping
Carbon Neutrality Target (0-1)	Sustainability	Public sources web scraping

Additional demographic and geographic variables were also collected to capture the broader institutional and national context in which each university operates. The most relevant variables include country, primary research topic, country GDP, GDP per capita, number of associated and child institutions, top collaborating country, university age, native English-speaking country, and whether the institution is private or not. Most of these variables were obtained from QS/THE institutional profiles or public databases such as OpenAlex and ROR, while some variables were completed through public web scraping.

All these datasets used throughout the thesis are available in the main project github repository, together with the code and supporting files used. The repository can be accessed through the following link: <https://github.com/arnaucanela11/upf-rankings>.

3.3 Performance of UPF

To better understand UPF's current position, the following figures present the performance across all pillars in the latest editions of the QS and THE rankings. The first set of charts

(left) reports UPF's score in each pillar, together with the pillar's weight in the overall ranking. Since all pillar scores are normalized on a 0–100 scale, they are directly comparable within each ranking. The second set of charts (right) translates these scores into nominal ranking points. For each pillar, UPF's contribution to the overall score is calculated as the pillar score multiplied by its weight. This is then compared with the maximum number of points that could be obtained from that pillar. The gap between the two bars therefore represents the number of points that UPF is failing to capture.

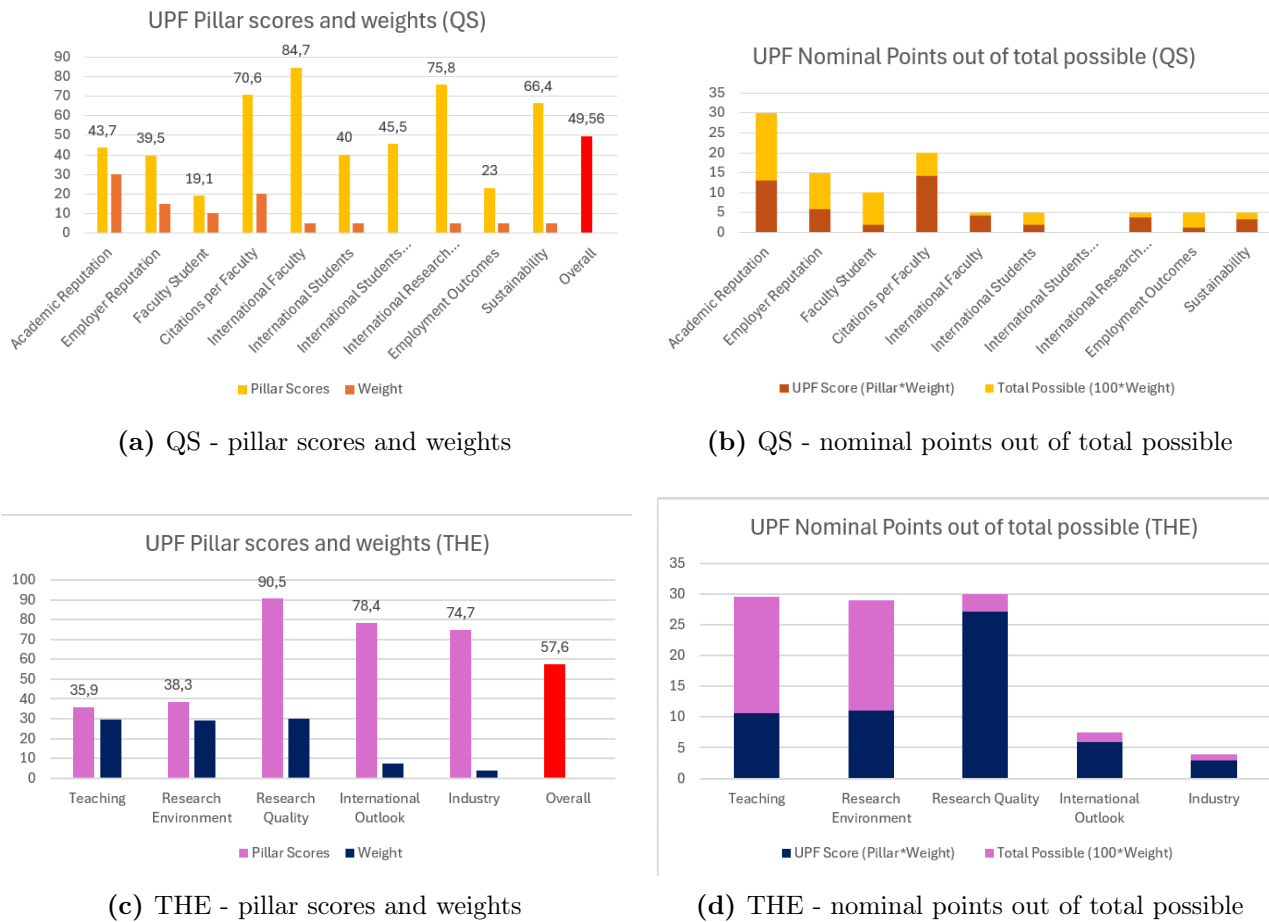


Figure 3: UPF's pillar performance in QS (top) and THE (bottom). Left: pillar scores together with each pillar's weight. Right: nominal points captured versus the maximum attainable; the gap is the number of points UPF fails to capture.

The results reveal a consistent pattern across both rankings: UPF underperforms in some of the most heavily weighted pillars. In QS, the largest point losses occur in pillars that measure the teaching quality (“Academic Reputation” and “Faculty-Student Ratio”) and in “Employer Reputation”, while in THE they occur in “Teaching” and “Research Environment”. Because these pillars carry substantial weights, even moderate improvements could have a significant impact on the overall score.

By contrast, UPF performs exceptionally well in research-related indicators that measure impact rather than scale. It scores highly in “Research Quality” (THE) and “Citations per Faculty” (QS), capturing most of the available points in these pillars. This suggests that UPF's research output is highly influential relative to its size.

A particularly interesting contrast emerges within THE. While UPF achieves an excellent score in “Research Quality”, it performs considerably worse in “Research Environment”. This indicates

that the university produces highly impactful research but does so with a relatively smaller research base. In other words, UPF appears to excel in the quality of its research rather than in its volume or resource intensity. From a strategic perspective, expanding research capacity while maintaining current quality levels could generate substantial gains in ranking performance.

Finally, QS highlights a weakness in employer-related dimensions. UPF obtains relatively low scores in “Employer Reputation” and “Employment Outcomes”, suggesting that its labour market recognition lags behind its academic and research performance. This pattern is less visible in THE, where the university performs relatively well in “Industry”, although this pillar carries only a small weight in the overall ranking.

3.4 Descriptive evidence

The next step is to investigate the composition of each pillar. To do that, each pillar score is plotted against the university’s ranking position rather than its overall score. This provides a direct view of how individual indicators are related to the final ranking and allows the identification of non-linearities, dispersion patterns, and potential differences across ranking systems. To make it more visually understandable, each dot represents the average of ten universities, grouped by the order of the ranking. That is, the first 10 universities represent a single dot, from top 10 to top 20 another dot and so on. In all figures, Universitat Pompeu Fabra (UPF) is highlighted to facilitate comparisons with the broader distribution of institutions.

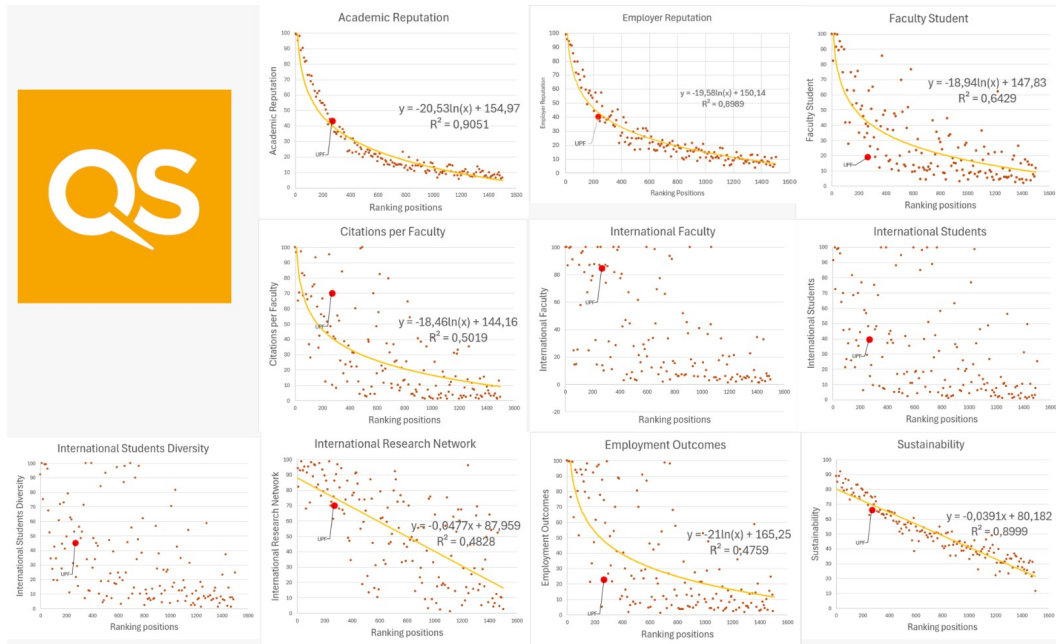


Figure 4: QS pillar scores against ranking position (each dot is a 10-university average), with UPF highlighted.

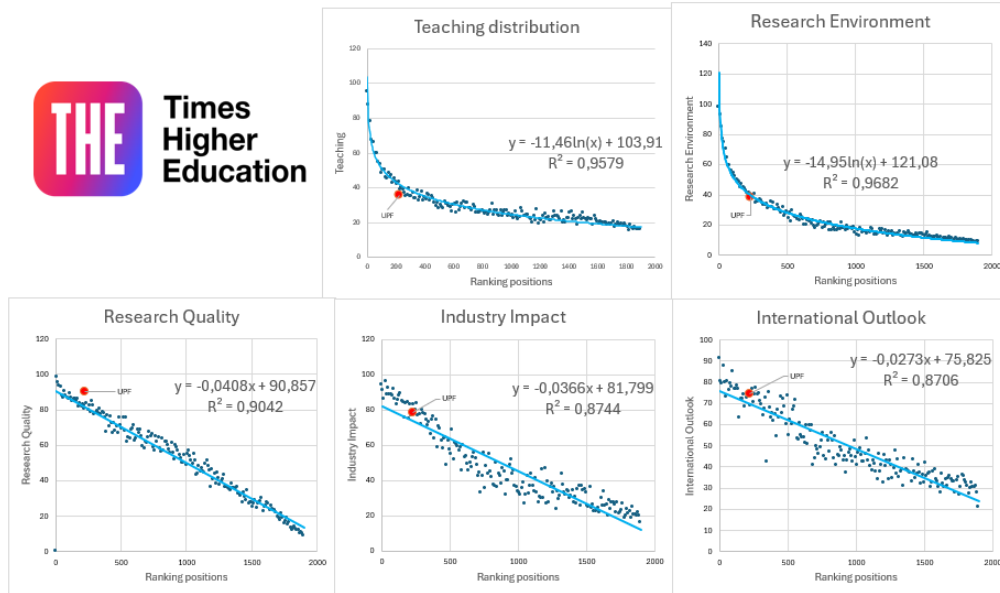


Figure 5: THE pillar scores against ranking position, with UPF highlighted.

An important finding from these plots is that the relationship between pillar scores and ranking position is not always linear. In particular, Academic Reputation (QS), Teaching (THE), and Research Environment (THE), show a strongly convex pattern. This means that a small group of top universities obtain very high scores, while the majority of institutions are concentrated much lower in the distribution. As a result, even universities with reasonably good performance can remain far from the maximum score. An important finding is that these 3 convex pillars are the same where UPF misses the largest number of points: therefore, part of UPF’s gap in these pillars may not be a particular problem of our university, but a consequence of the way the indicators are constructed.

Other indicators display a more linear relationship with ranking position. This is particularly evident for “Sustainability” (QS) and “Research Quality” (THE), where changes in the pillar score are associated with relatively constant changes in ranking position across the distribution.

Beyond the convexity of the relationships, the plots reveal another important difference between both rankings, the dispersion of indicators. In THE, most pillars show relatively little dispersion, meaning that universities with similar ranking positions tend to obtain similar scores across the pillars. In contrast, QS indicators are much more dispersed. While “Academic Reputation” (QS), “Employer Reputation” (QS) and “Sustainability” (QS) remain concentrated among the trend, all the other variables vary substantially among universities with similar overall ranks. This suggests that THE is driven more strongly by a common underlying dimension of institutional performance, whereas QS combines a broader set of partially independent characteristics, allowing universities to achieve similar ranking positions through different combinations of strengths and weaknesses. This finding motivates the following principal components analysis.

4 Unpacking QS and THE: latent dimensions and institutional clusters

The first empirical question is simple: do QS and THE actually measure different things? Publishing different pillars does not, in itself, mean that those indicators capture different

forms of excellence. If all the indicators move together, then the apparent multidimensionality is cosmetic: behind the multiple pillars there is really only one aspect that the ranking is capturing, and universities that are strong on one dimension are strong on all the others. If, on the other hand, the indicators separate universities into distinct profiles, then the ranking is picking up different routes to visibility.

Two tools are used. Principal component analysis (PCA) examines whether the pillar indicators are independent dimensions or whether they collapse into a single underlying factor. Cluster analysis then groups universities by the shape of their indicator profile, rather than by their level, creating different groups or clusters of similar universities. Full PCA and clustering outputs are reported in Appendix A.2.

4.1 Principal Components Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms a large set of correlated variables into a smaller set of uncorrelated variables called principal components. It accomplishes this by mathematically rotating the data space to project the maximum possible variance onto the first few orthogonal axes, effectively filtering out noise and data redundancy.

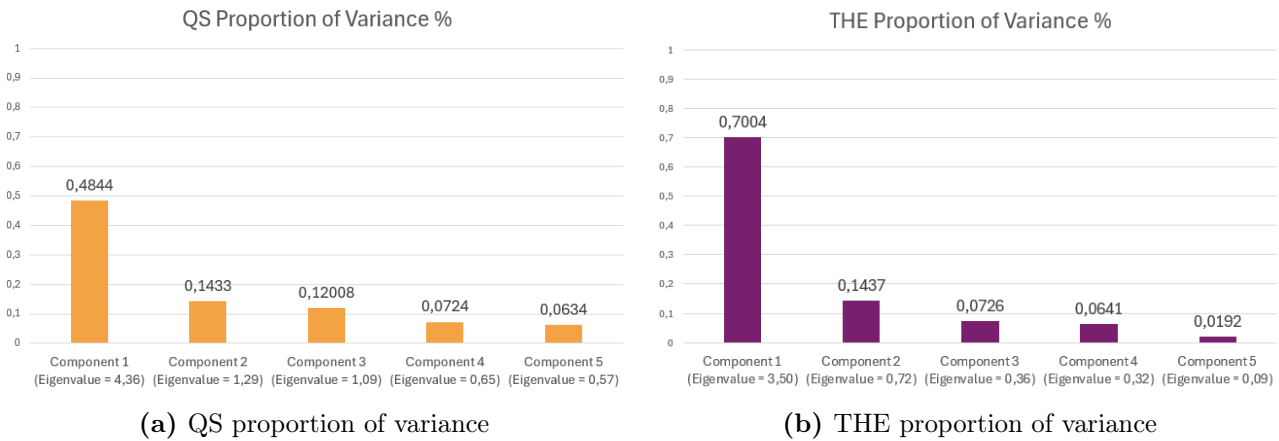


Figure 6: Proportion of variance explained by each principal component in QS and THE, with eigenvalues.

For clearance, the first component in THE is composed of: a linear combination of all five standardized ranking pillars, where each variable contributes positively and with remarkably balanced weights, specifically loading at 0.4541 for Teaching, 0.5014 for Research Environment, 0.4302 for Research Quality, 0.4566 for Industry Impact, and 0.3859 for International Outlook.

Table 4: Latent structure of THE and QS.

Ranking	Variance explained by PC1	Variance explained by PC1–PC2	Components with eigenvalue > 1	Main interpretation
THE	70.04%	84.41%	1	Dominated by one underlying characteristic of institutional strength
QS	48.40%	62.70%	3	More multidimensional, with several partially distinct performance dimensions

In the case of THE, the first principal component explains 70.04% of the total variance across the five pillars. Moreover, all five pillars load strongly and positively on the first component: Teaching, Research Environment, Research Quality, Industry Impact and International Outlook all move in the same direction. In practical terms, THE does not appear to measure five fully independent dimensions of excellence. Instead, its pillars largely behave as different expressions of the same underlying institutional capacity.

This structure contrasts with the QS results. In QS, the first component explains only 48.4% of the variance, and three components are needed to capture the main structure of the data according to the eigenvalue criterion. This indicates that QS is more multidimensional: reputation, employability, internationalisation, research connectivity, sustainability and faculty-related indicators do not all collapse into one single axis.

Therefore, the PCA results suggest that THE behaves primarily as a single underlying characteristic of accumulated institutional strength. To perform well in THE, a university generally needs to be strong across several pillars simultaneously. QS, on the other hand, allows for more differentiated institutional profiles, because its indicators capture several partially independent dimensions of university performance.

For the subsequent clustering analysis, the first two principal components are retained in both cases. Although the first component is the dominant one in THE, using two components allows the clustering analysis to preserve most of the variance in the data while also capturing the secondary differences between institutional profiles.

4.2 Clustering

The clustering results reinforce the conclusions obtained from the PCA. In the case of THE, the clusters are mainly separated by their overall level of performance rather than by clearly distinct institutional models.

For instance, Cluster 1 represents the strongest group of universities, with very high scores across Teaching, Research Environment, Research Quality, Industry Impact and International Outlook. This cluster can be interpreted as the elite group of research-intensive institutions.

Cluster 2 also contains strong universities, but with slightly less dominant performance than

Cluster 1. These institutions remain broadly balanced across the five pillars, suggesting a strong all-round profile rather than a narrow specialisation. Cluster 3 is the lowest-performing group, with scores below the sample average across all THE pillars. Finally, Cluster 4 occupies a more intermediate position and is relatively more associated with International Outlook and Research Quality than with Teaching, Research Environment or Industry Impact.

The QS clustering results show a more differentiated structure. While one cluster contains relatively lower-performing universities across most indicators (Cluster 1), the remaining clusters reveal more distinct institutional profiles. One group is mainly characterised by internationalisation, with stronger International Faculty Ratio and International Student Ratio scores but weaker reputation and employability indicators (Cluster 2). Another group is defined by stronger Academic Reputation, Employer Reputation, Employment Outcomes and research-related performance (Cluster 3). The final group represents global all-rounders, with strong scores across reputation, employability, research connectivity, internationalisation and sustainability (Cluster 4).

As a result, QS allows for more diverse institutional profiles. However, the clustering results also show that specialisation alone is not enough to define the strongest universities. In both rankings, the top-performing clusters are not those that excel in only one dimension, but those that combine several advantages simultaneously. In fact, the highest-ranked institutions in both systems remain those with broad and cumulative strengths across multiple dimensions.

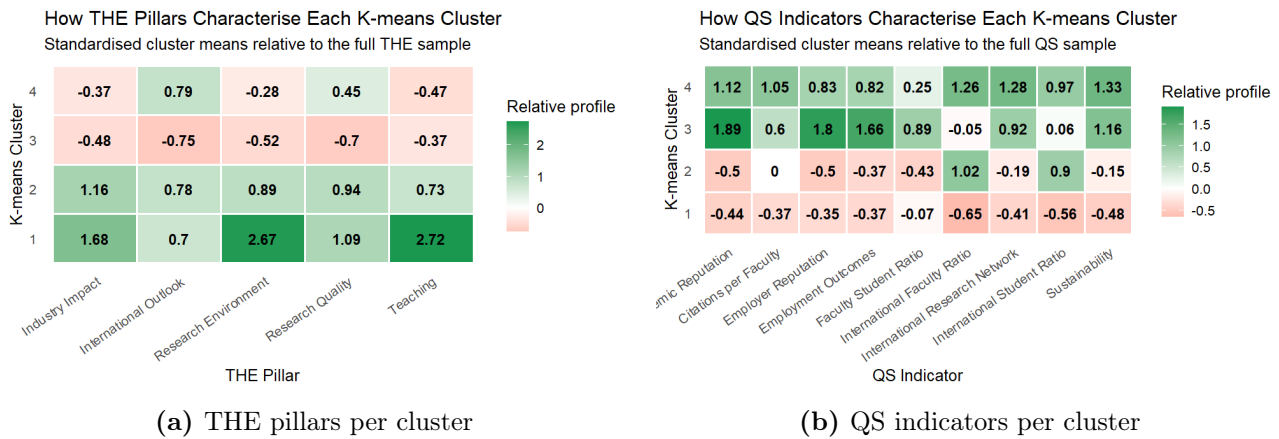


Figure 7: How THE pillars (left) and QS indicators (right) characterise each k-means cluster. Cells are standardised cluster means relative to the full sample.

4.3 UPF's diagnosis

The clustering results place Universitat Pompeu Fabra in Cluster 2 in both THE and QS. However, this does not mean that UPF has the same profile in both rankings. Since the two rankings are built around different indicators and latent structures, belonging to Cluster 2 has a different substantive meaning in each case.

In THE, Cluster 2 represents a group of strong and broadly balanced universities. These institutions perform above the sample average across all five pillars: Teaching, Research Environment, Research Quality, Industry Impact and International Outlook. UPF's position in this cluster suggests that, according to THE, it belongs to a group of universities with solid overall institutional performance, but still below the most extreme elite cluster.

In QS, Cluster 2 has a different interpretation. This cluster is mainly characterised by strong internationalisation, especially in International Faculty Ratio and International Student Ratio, but weaker performance in reputation-based and employability-related indicators such as Academic Reputation, Employer Reputation and Employment Outcomes. UPF’s presence in this cluster suggests that its international profile is relatively strong, but that this strength has not fully translated into global reputation or employer recognition.

The comparison therefore reveals an important diagnostic difference. THE portrays UPF as part of a strong, balanced group of universities that perform well across the main institutional pillars, although not at the level of the global elite. QS, by contrast, highlights a more specific strategic constraint: UPF appears internationally connected, but its global reputation and employer visibility remain comparatively weaker.

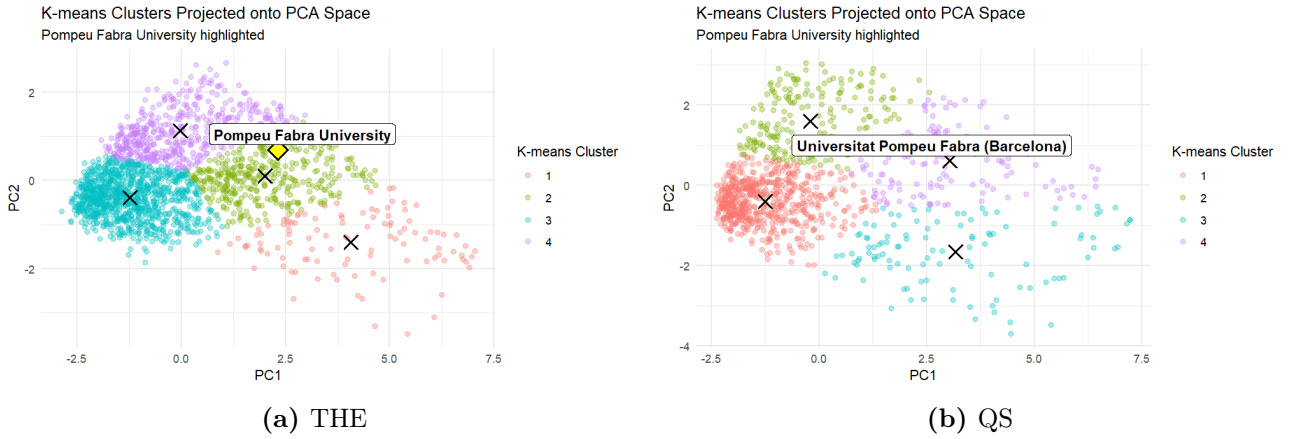


Figure 8: K-means clusters projected onto PCA space, with Pompeu Fabra University highlighted, for THE (left) and QS (right).

5 Robustness of QS Methodology

The previous section showed that QS and THE reward different models of excellence. This raises a natural second question: how robust is the methodology of these rankings? If the weights attached to each of the pillars changed, would universities stay where they are, or would the ranking rearrange itself?

QS is the natural first case for this kind of test, for two reasons. First, the previous section showed that QS is genuinely multidimensional, so changing the relative importance of its signals can plausibly redistribute the ranking. THE is more dominated by a single dimension and is therefore less exposed to weight perturbation. Second, QS has actively redesigned its methodology in recent years, adding Sustainability, Employment Outcomes and International Research Network as new pillars. The question of how much a ranking moves when its weights change is therefore not hypothetical: QS itself has been changing them.

5.1 Monte Carlo simulations

The analysis uses the cleaned QS 2024 dataset, containing 1,085 universities with complete information for all nine pillars. To assess the sensitivity of the ranking to changes in indicator weights, a Monte Carlo simulation is performed. In each of 1,000 simulations, the pillar weights

are randomly perturbed using a uniform distribution within $\pm 10\%$ of their baseline values and then rescaled to sum to one. New university scores and ranking positions are calculated and arranged for each scenario, generating over one million simulated rank observations. For each university, the analysis records the average rank, best and worst rank achieved, rank range, and rank standard deviation. Additional $\pm 5\%$ and $\pm 20\%$ perturbation scenarios are used as robustness checks. Further methodological details are provided in Appendix A.4.

The $\pm 10\%$ simulation shows that QS is neither arbitrary nor perfectly precise. Across the 1,085 universities, the average rank range is 42.1 positions, and the average standard deviation of the simulated rank is 7.6 places. The most sensitive university moves across 111 positions depending on the weighting scheme.

Table 5: QS rank uncertainty under Monte Carlo weight perturbation.

Perturbation scenario	Average rank range	Average rank SD	Maximum rank range	Maximum rank SD
$\pm 5\%$	21.71	3.91	67	13.26
$\pm 10\%$	42.11	7.58	111	25.47
$\pm 20\%$	82.82	14.84	213	44.86

The three perturbation scenarios scale as expected. When the admissible weight perturbation doubles from $\pm 5\%$ to $\pm 10\%$, the average rank range almost doubles. When it doubles again to $\pm 20\%$, uncertainty widens further. The pattern is therefore not an artefact of a particular choice of perturbation size: larger methodological changes generate proportionally larger positional uncertainty.

The most important result, however, is not the average level of uncertainty. It is how that uncertainty is distributed across the ranking:

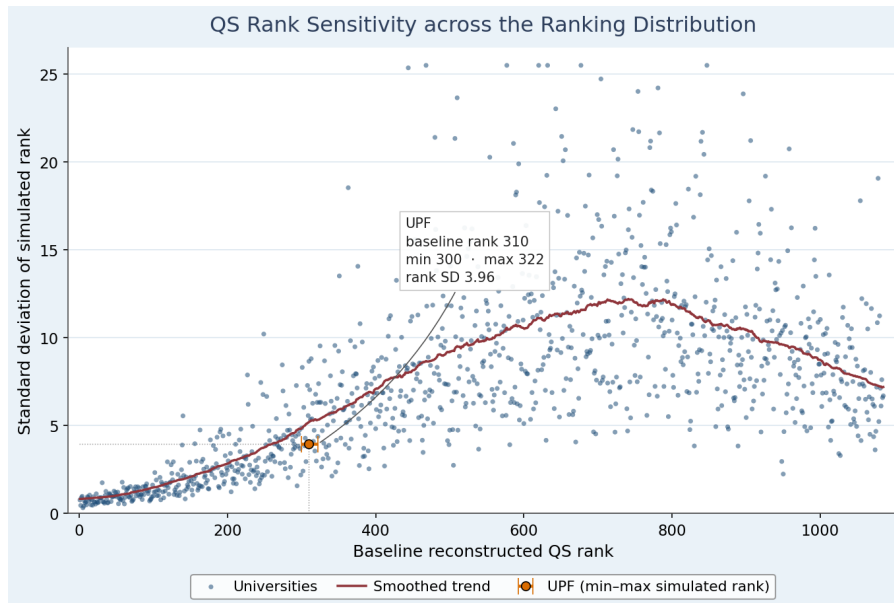


Figure 9: QS rank sensitivity across the ranking distribution under $\pm 10\%$ weight perturbation, with UPF highlighted.

The figure shows a clear non-linear pattern. At the very top of the ranking, simulated rank volatility is close to zero: top universities barely move when the weights change. Sensitivity

then rises sharply through the middle and lower-middle of the distribution, reaching its peak somewhere between baseline ranks 500 and 800, before declining again at the very bottom.

This pattern follows directly from the cluster analysis in Section 4.2. The global QS elite is balanced across most indicators, so modest changes in weights cannot easily displace it. Universities in the middle of the ranking are more compressed (many universities clustered within a narrow score range) and their profiles are more uneven (strong in some indicators, weak in others). When the relative value of those indicators changes, the ordering among these universities shifts. The lower tail of the ranking is also relatively stable, but for the opposite reason: these universities score uniformly low across indicators, so reweighting cannot rescue them.

5.2 The UPF case

In the QS baseline ranking, UPF is positioned at rank 310. Under the main $\pm 10\%$ Monte Carlo specification, its simulated rank varies only between 300 and 322, with a rank range of 22 positions and a simulated rank standard deviation of 3.96. Both values are well below the sample averages of 42.1 and 7.58 reported in the previous subsection. UPF is therefore noticeably more stable than the typical university under moderate weight perturbation.

Table 6: UPF rank uncertainty under Monte Carlo simulations.

Scenario	Mean simu- lated rank	Best rank	Worst rank	Rank range	Rank SD
$\pm 5\%$	309.88	305	317	12	2.10
$\pm 10\%$	310.53	300	322	22	3.96
$\pm 20\%$	312.09	291	338	47	7.54

The stress tests reinforce the same picture. Even under the strongest $\pm 20\%$ perturbation, UPF’s simulated rank only ranges from 291 to 338. Across the full set of 1,000 simulations in the main scenario, UPF never enters the top 200 and never falls outside the top 400. Moderate changes in the weights cannot move UPF substantially upward, because its imbalance relative to the QS elite is structural rather than methodological. To put it differently: UPF is not held back by a particular weighting choice. It is held back by an indicator profile that no plausible weighting scheme would translate into elite status.

5.3 Adding new pillars

Over the last few years, QS has introduced several new indicators, including International Research Network, Employment Outcomes, Sustainability, and International Student Diversity. As a result, changes in a university’s ranking position may reflect not only changes in performance but also changes in the ranking methodology itself.

To isolate the effect of these methodological revisions, a counterfactual ranking was constructed

for each edition using a fixed set of indicators and weights corresponding to the original QS methodology from 2017: “Academic Reputation”, “Employer Reputation”, “Faculty-Student Ratio”, “Citations per Faculty”, “International Faculty Ratio”, and “International Student Ratio”. Universities were then re-ranked using this constant methodology and compared with the official QS rankings.

The similarity between the official and reconstructed rankings was measured using correlations:

Table 7: Correlation between official and reconstructed QS rankings, by edition.

QS Edition	Pillar added	Correlation
2023	International Research Network and Employment Outcomes added	0.987
2024	Sustainability added	0.967
2025	-	0.965
2026	International Student Diversity added	0.960

The correlations remain above 0.96 in all years, indicating that methodological changes have had a limited impact on the overall structure of the ranking. The relative ordering of universities remains largely unchanged even after the introduction of new indicators. The same conclusion can be drawn: changes in the methodology can’t help UPF to climb positions.

6 Ranking mobility

The previous section examined how sensitive rankings are to methodological changes. This section addresses a different question: holding the methodology fixed, how difficult is it for a university to move up the ranking?

The analysis uses THE 2024 because it publishes an overall score for the full sample of 1,904 universities, with no missing values on the variables required. For each university, four measures are constructed.

1. **Score gap to move up 10, 25 or 50 analytical positions:** the increase in overall score needed to reach the university that currently sits 10, 25 or 50 places above.
2. **The score cushion before falling 10, 25 or 50 places:** the score distance to the university below.
3. **Local score gap:** defined as the average distance to the immediate neighbour above and below; this captures how dense a university’s local ranking environment is.
4. **Rank return per score point:** calculated as 10 divided by the score gap to move up 10 places. This measures how many ranking places one additional THE score point is worth around each university.

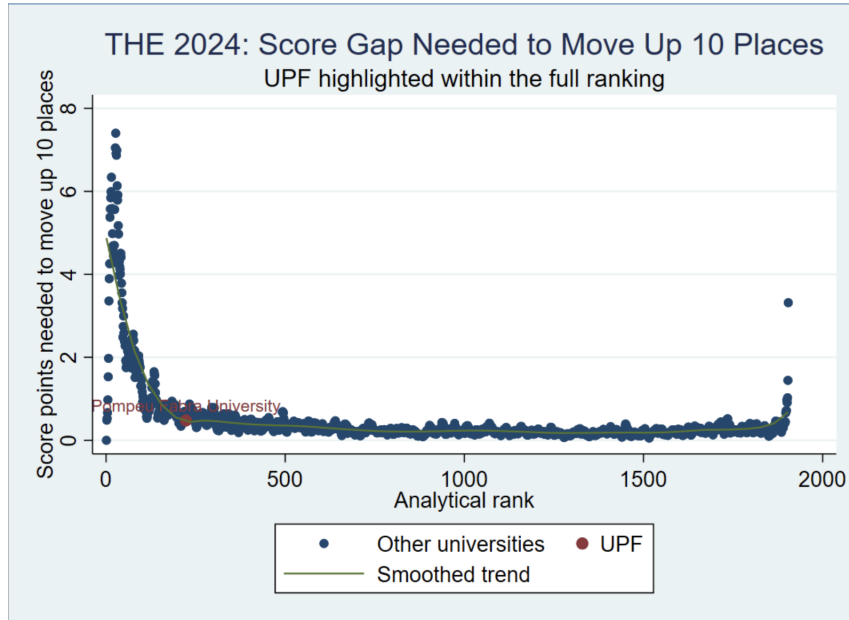
Table 8 reports the four mobility measures averaged by rank band. The full version, including the 25- and 50-place gaps, is in Appendix A.

Table 8: Mobility structure of THE 2024 by rank band.

Rank band	Gap to move up 10 places	Cushion before falling 10 places	Local score gap	Rank re-turn per score point
Top 50	4.30	4.36	0.458	3.51
51 to 100	1.98	1.74	0.183	5.20
101 to 200	0.81	0.77	0.080	13.21
201 to 500	0.45	0.44	0.044	24.60
501 to 1000	0.25	0.25	0.025	44.30
1001+	0.23	0.25	0.024	50.80

The results show that ranking mobility is highly non-linear. Universities at the top of the ranking are separated by large score differences, while those further down are much more densely packed. On average, a Top 50 university needs 4.3 additional score points to gain 10 positions, compared with only 0.25 points for a university ranked between 501 and 1000. Consequently, one additional score point is worth only around 3.5 positions in the Top 50, but more than 40 positions in the lower part of the ranking.

Figure below illustrates this pattern across the full distribution. The score required to move up increases sharply near the top, making elite positions structurally harder to reach. Middle-ranked universities face the opposite situation: ranking positions are easier to gain, but also easier to lose with small score changes.

**Figure 10:** Score gap required to move up 10 places in THE 2024, with UPF highlighted within the full ranking.

6.1 UPF position

UPF is ranked 223rd in THE 2024, placing it in the 201–500 band. Its mobility indicators are close to the average for this group. A gain of 0.48 score points would be sufficient to move up 10 positions, but the required improvement rises to 1.26 points for 25 positions and 3.18 points for 50 positions. This shows that larger jumps become increasingly difficult.

At the same time, UPF’s position is relatively stable. It can lose 0.88 points before falling 10 places, compared with a band average of 0.44, and its local score gap is more than twice the average. UPF is not located in a particularly crowded part of the ranking and is less exposed to small fluctuations than comparable universities.

Table 9: UPF within the THE 2024 mobility structure.

Measure	UPF	201 to 500 band average
Gap to move up 10 places	0.48	0.45
Gap to move up 25 places	1.26	1.14
Gap to move up 50 places	3.18	2.35
Cushion before falling 10 places	0.88	0.44
Local score gap	0.101	0.044
Rank return per score point	20.90	24.60

6.2 Are UPF’s required gains historically plausible?

The previous analysis quantified the score improvements required for UPF to move up the ranking. The next question is whether gains of this magnitude are historically realistic.

Using the THE panel from 2016–2024, annual score changes were calculated for all universities appearing in consecutive editions. Because THE scores tend to increase over time as new universities enter the ranking, raw score changes overstate genuine progress. To account for this, annual changes were adjusted by subtracting the average score increase in each year, capturing improvement relative to the rest of the system.

The results suggest that modest ranking gains are realistic. A score increase of 0.48 points, sufficient to move UPF up 10 places, was achieved by 36.6% of all universities in a typical year. Moving up 25 places requires a larger improvement and was achieved by 24.8% of universities. By contrast, the 3.18-point gain needed to move up 50 places was observed in only 9.3% of university-years.

Overall, the evidence suggests that incremental progress in THE is achievable, whereas large jumps are uncommon. Combined with the cluster and mobility analyses, the results indicate that UPF’s position is relatively stable and that advancing towards higher tiers of the ranking is likely to require sustained improvements over several years rather than a single exceptional performance.

Table 10 compares UPF’s mobility thresholds with the share of historical changes, both raw and adjusted, that would have been sufficient to reach them.

Table 10: Historical plausibility of UPF’s THE mobility gaps.

UPF mobility target	Score gain re-quired	Raw share of all annual changes	Adjusted share of all annual changes	Raw share of positive gains	Adjusted share of positive gains
Move up 10 places	0.48	61.09%	36.61%	86.03%	79.15%
Move up 25 places	1.26	45.03%	24.84%	63.42%	53.71%
Move up 50 places	3.18	20.67%	9.34%	29.10%	20.19%

7 Proxy Variables: Final Actionable Analysis

The previous sections analysed university rankings from the perspective of their published pillars. This section takes a different approach. Instead of studying only the official pillar scores published by QS and Times Higher Education, it builds a set of observable proxy variables that approximate the underlying sub-pillars used by the rankings. The motivation for doing this is that ranking providers publish final positions, overall scores and broad pillar-level indicators, but many of the underlying raw inputs and transformations are not fully transparent. As a result, universities can observe where they perform well or poorly, but they cannot always identify which concrete institutional variables are most strongly associated with ranking outcomes. Proxy variables aim to help bridging this gap.

The analysis uses universities included in the THE 2024 ranking. Observations with missing values in any variable used in the corresponding model were removed, and extreme outliers were excluded. After cleaning, the modelling sample contains approximately 1,500 universities. See all variables used and its sources in the Table 3 (section 3.2).

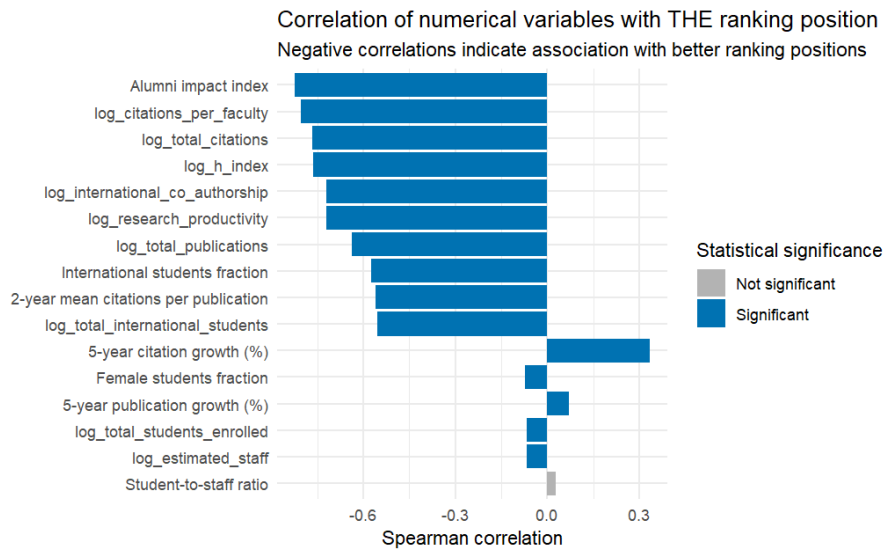


Figure 11: Spearman correlation of numerical proxy variables with THE ranking position. Negative correlations indicate association with better ranking positions.

Before moving to the predictive models, the proxy variables were first examined descriptively. The correlation analysis also reveals a clear pattern: the strongest associations with better THE ranking positions come from research impact and visibility variables, particularly alumni impact, citations per faculty, total citations, H-index, international co-authorship and research productivity.

The modelling process then compared increasingly flexible specifications. The first benchmark was a standard linear regression including all retained proxy variables. This model is easy to interpret and provides a baseline for understanding whether the relationship between proxy variables and ranking position is approximately linear. Two stepwise alternatives were then estimated using AIC and BIC. StepAIC searches for a better trade-off between fit and complexity, while StepBIC applies a stronger penalty for model size.

After the linear models, a Generalized Additive Model (GAM) was estimated. The motivation for using a GAM is that several proxy variables are unlikely to affect ranking position in a strictly linear way. For example, the marginal ranking effect of increasing research productivity or H-index may be larger at lower levels and smaller once a university is already highly research-intensive.

Finally, two machine-learning models were tested: Random Forest and Neural Network. Random Forest is useful because it can capture non-linear effects and interactions without specifying them manually. All models were evaluated using an 80/20 train-test split and 10-fold cross-validation on the training sample. Performance was compared using RMSE, MAE and R^2 .

Table 11: Predictive performance of the models tested.

Model Type	CV RMSE	Test RMSE	Test MAE	Test R^2
Initial Linear Model	251.86	239.30	189.21	0.78
Step BIC	251.68	239.45	189.43	0.78
Step AIC	252.04	241.61	191.02	0.78
GAM Model	228.73	213.50	167.16	0.83
Random Forest	218.44	192.33	145.06	0.86
Neural Network	258.73	227.12	178.46	0.80

The results show that the non-linear models perform better than the linear benchmarks. In particular, Random Forest achieves the best predictive performance (low RMSE and high R^2), while the GAM also performs strongly and offers a better balance between accuracy and interpretability. This level of predictive performance means that the model explains around 86% of the variation in THE ranking positions in the test set. In practical terms, its average absolute error is approximately 145 ranking positions, meaning that, on average, the predicted rank differs from the real rank by about 145 places. This is still a considerable margin, but it is the lowest error among all models tested. The linear models remain useful as transparent benchmarks, but they are consistently less accurate.

Finally, a second modelling exercise added geographic and demographic variables (dataset 2), including country, top collaborating country, primary research topic, GDP, native English-speaking country, etc. However, these variables did not consistently improve predictive performance. Although they provide useful contextual information, they increased model complexity

and introduced additional noise.

These final linear models, together with Random Forest and GAM, were included in the interactive tool as alternative model choices for creating new scenarios with the available data. Ultimately, this was the main purpose of the final modelling stage: to provide models that allow users to experiment with actionable variables. The interactive tool is where this scenario analysis can be carried out.

8 Conclusions

As shown throughout this thesis, the QS and Times Higher Education rankings are calculated as a weighted sum of several pillars, where each pillar accounts for a different characteristic of a university and gives more weight to some dimensions than to others. A first assessment of how UPF performs in these rankings reveals that the university loses most of its available points in the pillars that carry the largest weights: Academic Reputation in QS, and Teaching and Research Environment in THE. At first sight this looks like a clear sign of underperformance. However, these same pillars follow a strongly convex distribution, which makes a high score structurally difficult to reach. Missing points in them is therefore a very common situation for the majority of universities, since the top scores are reserved for the global elite and the cost of every additional point grows almost exponentially.

It is also worth highlighting that UPF obtains an almost perfect score in Research Quality (THE), a pillar that carries a high weight (30%) and follows a roughly linear distribution. This contrasts sharply with its much weaker performance in Research Environment (THE). The combination tells a consistent story: UPF produces highly impactful research, but does so with a comparatively small research base. Even taking into account the convex boundary of some pillars, this suggests that UPF should try to enlarge the institutional size of its research (number of papers, funding, research staff) while preserving the high quality it already achieves.

A Principal Components Analysis was then carried out to understand how each system is structured internally, revealing a clear difference between the two. In THE, around 70% of the variance is explained by the first component, which is made up of a positive and very similar weighting of all five pillars. This means that THE essentially rewards a single underlying characteristic, so that all its pillars tend to move together and a university needs to be strong everywhere at once. QS, on the contrary, allows for a broader set of profiles that can lead to a good position, the most relevant being universities with high quality research and universities with a high degree of internationalisation.

The clustering exercise pointed in the same direction. In neither ranking does UPF belong to the elite cluster, the group of universities that score highly across all pillars. In THE it falls into a cluster of universities that perform fairly evenly across the pillars but remain one step behind the global elite, with somewhat lower scores. In QS it belongs instead to a cluster defined by a strong degree of internationalisation together with weaker performance in reputation and employability indicators. The same institution therefore receives two different diagnoses: balanced but not yet elite in THE, and internationally connected but less recognised in QS.

Monte Carlo simulations were then run, changing the weights assigned to each pillar, leading to the conclusion that the methodology of both rankings is robust. No dramatic changes in

positions were observed even under large shifts in the weights, and UPF only moved from rank 291 in its best scenario to 338 in its worst. Adding new pillars did not alter the ordering either. UPF therefore has neither an advantage nor a disadvantage built into the way the rankings are constructed, which means it cannot expect to climb simply because the methodology changes.

The ranking as a whole also displays a very convex relationship between the overall score and the position obtained, meaning that the closer a university is to the top, the more points it needs to gain a single place. UPF sits in an unfavourable part of this distribution. Looking at the score increase required to move up ten positions, the universities right in front of UPF are separated by large score differences, while those just behind it are packed closely together. In practice this means that UPF needs a substantial improvement to advance only a few positions, while a small decline would be enough to lose many. The same fact can also be read in a positive way, since UPF stands at the very top of its group of universities with similar scores.

Because the pillars rely on opaque definitions that universities cannot fully reconstruct, a prediction model was built to forecast the final score, and therefore the position, in the ranking. This was done using a new dataset of tangible variables that universities can observe directly, such as the student to staff ratio or the number of citations in their papers. Most of these variables turned out to be significant, and the best-performing model accounted for around 86% of the variance in ranking positions. It would therefore be useful for UPF to look at which tangible variables matter most and to assess its own performance in each of them. For that purpose an interactive tool was created that lets the user change each variable and see, in real time, how the ranking position responds.

As a final recommendation, the advisory services offered by ranking providers such as QS could be valuable for UPF, given the limited transparency of some underlying variables and data collection procedures. This is especially relevant for indicators based on reputational surveys, as well as for key income-related variables that are difficult to reconstruct externally through proxy data. Nevertheless, this recommendation should be understood as a strategic possibility rather than a decision of this thesis: whether to engage with ranking providers is ultimately a decision that belongs to the university itself.

A Appendix

A.1 Methodological tables



Pillar / Subpillar	Weight	Description	Data source	Data availability
Teaching	29.5%	Measures the learning environment and institutional teaching capacity.		Yes (web scrapping)
Teaching Reputation	15%	Perceived teaching prestige among academics worldwide.	THE Academic Reputation Survey	No
Student Staff ratio	4.5%		Self-submitted institutional data + World Bank data	Yes (web scrapping)
Doctorate Bachelor ratio	2%		Self-submitted institutional data	No
Doctorate Staff ratio	5.5%		Self-submitted institutional data	No
Institutional Income	2.5%		Self-submitted institutional data + World Bank PPP	No
Research Environment	29%	Measures research volume, income, productivity and reputation.		Yes (web scrapping)
Research Reputation	18%	Perceived research prestige among academics worldwide.	THE Academic Reputation Survey	No
Research Income	5.5%		Self-submitted institutional data + World Bank PPP	No
Research Productivity	5.5%	Publications per academic/research staff member.	Elsevier/Scopus + self-submitted institutional staff data	No
Research Quality	30%	Measures citation impact, research excellence and influence.		Yes (web scrapping)
Citation Impact	15%	Average citation impact of publications.	Elsevier/Scopus	Partly
Research Strength	5%	75th percentile FWCI.	Elsevier/Scopus	No
Research Excellence	5%	Publications in the global top 10% by FWCI.	Elsevier/Scopus	No
Research Influence	5%	Influence of research based on the importance of the papers that cite it.	Elsevier/Scopus (calculated by THE)	No
International Outlook	7.5%	Measures internationalisation of students, staff and research collaboration.		Yes (web scrapping)
International Students	2.5%		Self-submitted institutional data + World Bank data	Yes (web scrapping)
International Staff	2.5%		Self-submitted institutional data	No
International Co-authorship	2.5%	Share of publications with at least one international co-author.	Elsevier/Scopus	No
Students Abroad	0%	Outbound exchange students relative to total students (not used in final score).	Self-submitted institutional data	No
Industry	4%	Measures knowledge transfer and links with industry.		Yes (web scrapping)
Industry Income	2%	Research income from industry per academic staff member.	Self-submitted institutional data + World Bank data	Partly
Patents	2%	Patents citing university research.	Elsevier/Scopus + patent-linked bibliometric data	No

Figure 12: THE World University Rankings: full pillar / sub-pillar methodology, weights, data sources and data availability.



Pillar / Subpillar	Weight	Description	Data source	Data availability
Academic Reputation	30%	Measures global academic prestige based on academics' nominations of excellent universities.	QS Academic Survey.	Only final scores
Employer Reputation	15%	Measures employers' perception of which universities produce strong graduates.	QS Employer Survey.	Only final scores
Faculty Student ratio	10%	Proxy for teaching capacity; faculty resources relative to student numbers.	QS-validated FTE faculty and FTE student data from institutions, agencies, web sources and third parties.	Only final scores
Citations per Faculty	20%	Measures research impact relative to academic staff size.	Elsevier/Scopus + validated faculty data	Only final scores
International Faculty ratio	5%	Share of foreign-national faculty among total faculty.	Institutional data submitted to QS.	Only final scores
International Student ratio	5%	Share of foreign-national students among total students.	Institutional data submitted to QS.	Only final scores
International Research Network	5%	Measures diversity of sustained international research partnerships.	Elsevier/Scopus	Only final scores
Employment Outcomes	5%	Measures graduate employability and alumni impact.	QS Graduate Employability data (own dataset)	Only final scores
Sustainability	5%	Measures sustainability performance of the institution.	QS Sustainability data (own dataset) + institutional data submitted to QS	Only final scores

Figure 13: QS World University Rankings: pillar methodology, weights, data sources and data availability.

A.2 Methodological description: PCA / Cluster analysis

```
. pca z_ar z_er z_fsr z_cpf z_ifr z_isr z_irn z_ger z_sus
```

```
Principal components/correlation      Number of obs   =    1,085
                                     Number of comp. =     9
                                     Trace              =     9
Rotation: (unrotated = principal)    Rho              =    1.0000
```

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	4.35983	3.06977	0.4844	0.4844
Comp2	1.29006	.202532	0.1433	0.6278
Comp3	1.08753	.436287	0.1208	0.7486
Comp4	.651244	.0806503	0.0724	0.8210
Comp5	.570593	.184406	0.0634	0.8844
Comp6	.386187	.0821622	0.0429	0.9273
Comp7	.304025	.0722044	0.0338	0.9611
Comp8	.231821	.113119	0.0258	0.9868
Comp9	.118702	.	0.0132	1.0000

```
Principal components (eigenvectors)
```

Variable	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6
z_ar	0.4274	-0.2651	-0.0080	-0.0812	0.0185	-0.2616
z_er	0.3617	-0.3421	0.1240	-0.4172	0.1453	-0.4940
z_fsr	0.1400	-0.3032	0.7166	0.5966	0.0328	0.0947
z_cpf	0.3050	0.1469	-0.3327	0.3511	0.7968	0.0283
z_ifr	0.2812	0.5804	0.2233	-0.1073	0.0033	0.1702
z_isr	0.2633	0.5443	0.3766	-0.1560	-0.0416	-0.2140
z_irn	0.3665	0.0412	-0.3123	0.3686	-0.4733	-0.1148
z_ger	0.3640	-0.2534	0.0464	-0.3740	0.0305	0.7607
z_sus	0.3984	0.0133	-0.2626	0.1690	-0.3404	0.1060

Figure 14: QS: principal components / correlation output (Stata), with eigenvalues, proportion of variance, and eigenvectors for the nine standardized QS indicators.

```
. pca z_Teaching z_Research_Environment z_Research_Quality z_Industry_Impact z_I
> nternational_Outlook
```

Principal components/correlation

Number of obs	=	1,904
Number of comp.	=	5
Trace	=	5
Rho	=	1.0000

Rotation: (unrotated = principal)

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	3.50177	2.78322	0.7004	0.7004
Comp2	.718544	.355325	0.1437	0.8441
Comp3	.363218	.0426496	0.0726	0.9167
Comp4	.320569	.224666	0.0641	0.9808
Comp5	.0959031	.	0.0192	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Comp4	Comp5
z_Teaching	0.4541	-0.4683	0.2576	-0.4603	0.5443
z_Research~t	0.5014	-0.2513	0.1356	-0.1279	-0.8067
z_Research~y	0.4302	0.4165	-0.7279	-0.3271	0.0672
z_Industry~t	0.4566	-0.2403	-0.2122	0.8067	0.1950
z_Internat~k	0.3859	0.6974	0.5834	0.1181	0.1019

Figure 15: THE: principal components / correlation output (Stata), with eigenvalues, proportion of variance, and eigenvectors for the five standardized THE pillars.

A.3 Methodological description: Proxy Analysis and Modeling

A.3.1 Exploratory Data Analysis (EDA)

Before estimating the models, the proxy dataset was cleaned and inspected to ensure that the final specification was consistent and not driven by data quality issues. First, all numerical variables were converted to a comparable numeric format, missing values were identified, and observations with missing information in the relevant variables were removed. Binary variables, such as the carbon neutrality target, were recoded as categorical indicators.

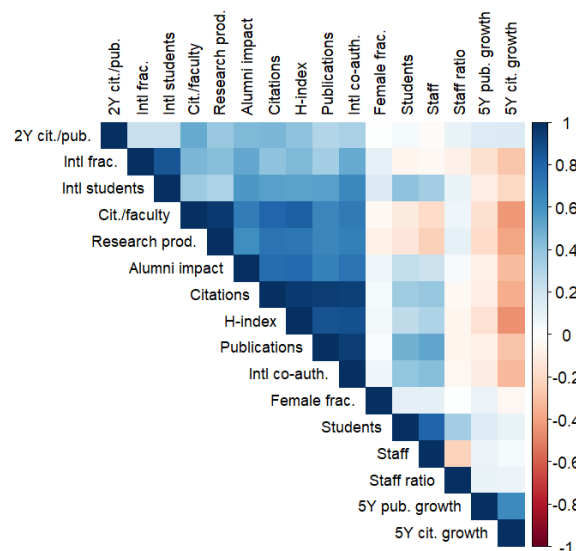


Figure 16: Correlation heatmap of the proxy variables used in the predictive analysis.

Finally, pairwise correlations were examined using Spearman correlations. This was done both to assess the relationship between each proxy variable and THE ranking position, and to detect potential multicollinearity between predictors. Highly overlapping variables were not all retained in the final model. Total students enrolled and estimated staff were excluded because they mainly capture institutional size. Total international students was also excluded because the international students fraction provides a cleaner size-adjusted measure of internationalisation. Finally, total citations and citations per faculty were excluded to avoid over-representing the same research-impact dimension already captured by H-index, publications, mean citations per publication and research productivity.

A.3.2 Model specification and evaluation

The dependent variable in all models was the THE world ranking position, where lower values indicate better performance. The final set of predictors combined institutional, research, internationalisation, employability and sustainability proxies.

The modelling sample was divided into a training set and a test set using an 80/20 split. Model selection and tuning were performed on the training data, using 10-fold cross-validation when required. Predictive performance was then evaluated on the test set using RMSE, MAE and R^2 .

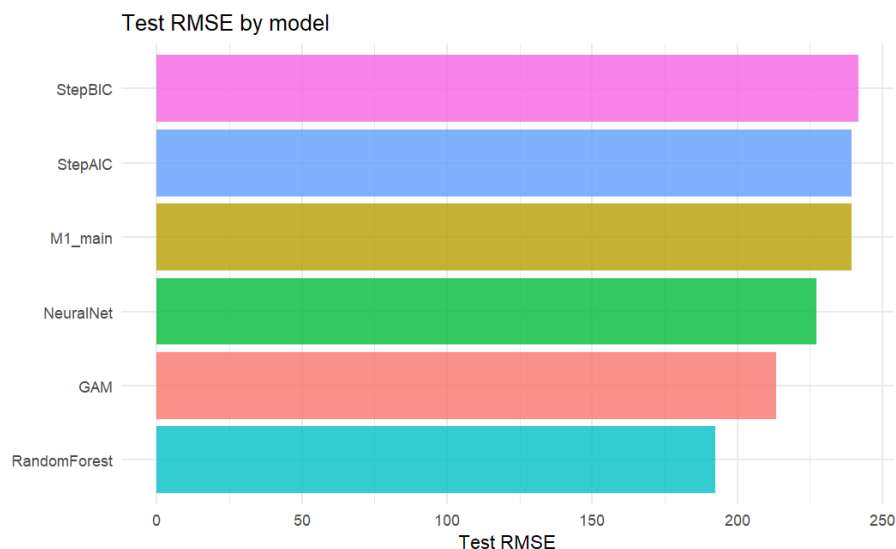


Figure 17: Test RMSE by predictive model. Lower RMSE values indicate better predictive performance.

Several model types were compared. First, a full linear regression was estimated as a transparent baseline. Two stepwise linear alternatives were then fitted using AIC and BIC criteria, allowing the model to reduce unnecessary predictors. Next, a Generalized Additive Model (GAM) was estimated to capture non-linear relationships while preserving interpretability. Finally, two machine-learning models were tested: Random Forest and Neural Network. The Random Forest model was tuned through cross-validation, while the Neural Network required dummy encoding and preprocessing of the input variables.

A.4 Methodological description: Monte Carlo sensitivity analysis (QS)

This appendix reports the supporting outputs for the QS 2024 Monte Carlo sensitivity analysis presented in Section 5.1. The analysis uses the 1,085 universities with complete information across the nine QS indicators and runs 1,000 simulations per perturbation scenario.

A.4.1 Data and simulation design

The official QS overall score was not used as the dependent object of the simulation because it is missing for many institutions outside the top-ranked segment, and the official rank variable includes tied positions and rank bands. Instead, a continuous QS-style score was reconstructed from the underlying indicator scores using the official QS weighting scheme. The reconstructed score is not treated as the official QS overall score; it is used as a transparent analytical benchmark from which sensitivity to weighting choices can be tested.

Table 12: Data and simulation design.

Item	Value
Ranking system	QS 2024
Original observations	1,168
Complete-indicator sample	1,085
Indicators used	AR, ER, FSR, CPF, IFR, ISR, IRN, GER, SUS
Official overall score used?	No
Official rank used as dependent variable?	No
Baseline score	Reconstructed weighted QS-style score
Main simulation	$\pm 10\%$ weight perturbation
Additional robustness scenarios	$\pm 5\%$ and $\pm 20\%$
Number of simulations per scenario	1,000

The baseline reconstructed score was calculated as:

$$\begin{aligned} \text{score} = & 0.30 \cdot \text{AR} + 0.15 \cdot \text{ER} + 0.10 \cdot \text{FSR} + 0.20 \cdot \text{CPF} \\ & + 0.05 \cdot \text{IFR} + 0.05 \cdot \text{ISR} + 0.05 \cdot \text{IRN} + 0.05 \cdot \text{GER} + 0.05 \cdot \text{SUS} \end{aligned}$$

A.4.2 Simulation procedure

In each simulation, the nine QS weights are randomly perturbed within a uniform range of $\pm 10\%$ relative to their baseline values, then normalised so that they sum to one. University scores are recalculated using these perturbed weights, and all universities are re-ranked. Repeating this 1,000 times for the 1,085 universities produces $1,085 \times 1,000 = 1,085,000$ simulated rank observations. For each university, the mean, best (minimum) and worst (maximum) simulated rank, the rank range and the rank standard deviation are computed.

A.4.3 Main $\pm 10\%$ uncertainty summary

Table 13: Summary of QS rank uncertainty under $\pm 10\%$ weight perturbation (1,085 universities).

Measure	Mean	Std. dev.	Min	Max
Mean simulated rank	543.00	313.23	1	1081.73
Best simulated rank	522.01	307.37	1	1072
Worst simulated rank	564.12	318.72	1	1085
Rank range	42.11	22.91	0	111
SD of simulated rank	7.58	4.48	0	25.47

A.4.4 Most stable institutions

Table 14: Most stable universities under $\pm 10\%$ weight perturbation.

Base rank	University	Country	Best rank	Worst rank	Range	SD
1	MIT	United States	1	1	0	0.00
2	University of Cambridge	United Kingdom	2	2	0	0.00
3	University of Oxford	United Kingdom	3	3	0	0.00
4	Harvard University	United States	4	4	0	0.00
7	ETH Zurich	Switzerland	7	7	0	0.00
5	Stanford University	United States	5	6	1	0.20
6	Imperial College London	United Kingdom	5	6	1	0.20
8	National Univ. of Singapore	Singapore	8	9	1	0.36
9	UCL	United Kingdom	8	9	1	0.36
32	University of Manchester	United Kingdom	32	35	3	0.43

A.4.5 Most sensitive institutions

Table 15: Most sensitive universities under $\pm 10\%$ weight perturbation.

Base rank	University	Country	Best rank	Worst rank	Range	SD
696	China Univ. of Petroleum	China	639	745	106	25.47
715	IIT Bhubaneswar	India	647	756	109	24.63
823	Beijing Univ. Posts & Telecom.	China	768	875	107	23.93
672	IIT Hyderabad	India	624	724	100	23.62
561	IIT (BHU) Varanasi	India	506	611	105	23.44
658	East China Univ. Sci. & Tech.	China	621	713	92	21.62
838	Univ. Central “Marta Abreu”	Cuba	783	883	100	21.60
809	Nanjing Univ. Aeronautics & Astro.	China	763	857	94	21.50
828	Nat. Research Saratov State Univ.	Russia	777	871	94	21.33
850	Huazhong Agricultural University	China	790	901	111	21.19

A.4.6 Robustness across perturbation scenarios

Table 16: Rank uncertainty across Monte Carlo perturbation scenarios, full sample and UPF.

Scenario	Avg. range	Avg. SD	Max range	Max SD	UPF range	UPF SD	UPF best	UPF worst	UPF mean
$\pm 5\%$	21.71	3.91	67	13.26	12	2.10	305	317	309.88
$\pm 10\%$	42.11	7.58	111	25.47	22	3.96	300	322	310.53
$\pm 20\%$	82.82	14.84	213	44.86	47	7.54	291	338	312.09

A.4.7 UPF indicator profile and probabilities

Table 17: UPF indicator profile in QS 2024.

University	AR	ER	FSR	CPF	IFR	ISR	IRN	GER	SUS
UPF (Barcelona)	30.9	22.8	13.1	55.6	77.9	26.6	38.4	15.3	13.2

Under the main $\pm 10\%$ scenario, UPF has a baseline reconstructed rank of 310 and a mean simulated rank of 310.53, varying between 300 and 322 (rank range of 22, SD of 3.96). Across the 1,000 simulations, UPF has a 0% probability of entering the top 100 or top 200, but a 100% probability of remaining within the top 500.

A.4.8 Supplementary figures

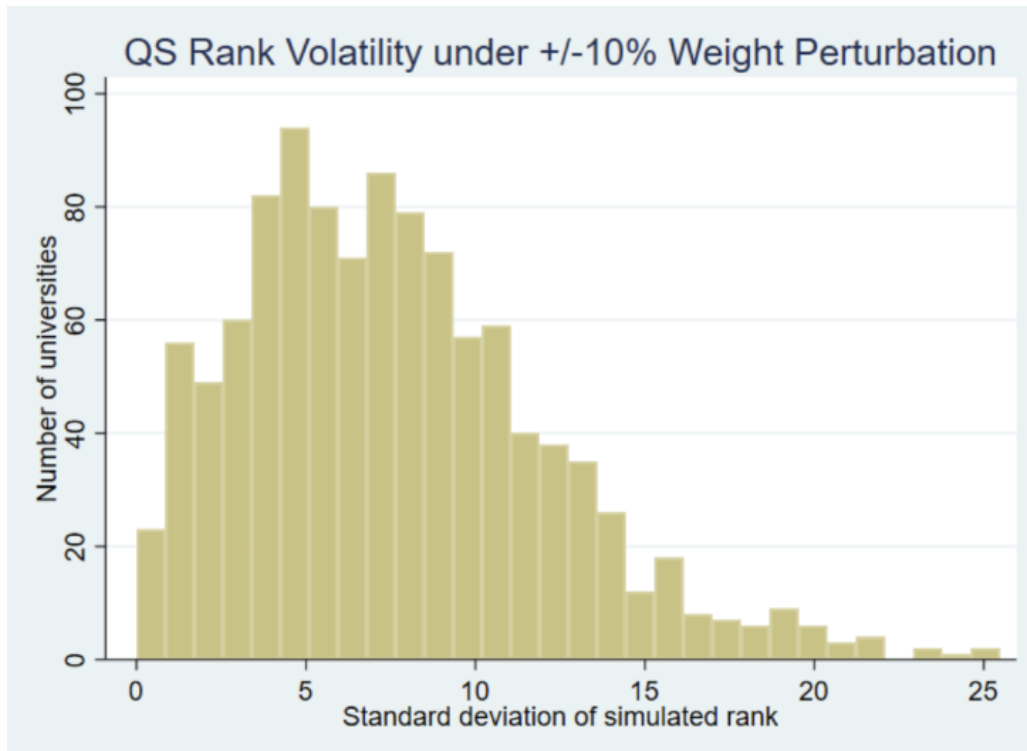


Figure 18: Distribution of QS rank volatility under $\pm 10\%$ weight perturbation. Rank volatility is measured as the standard deviation of each university's simulated rank across 1,000 Monte Carlo simulations.

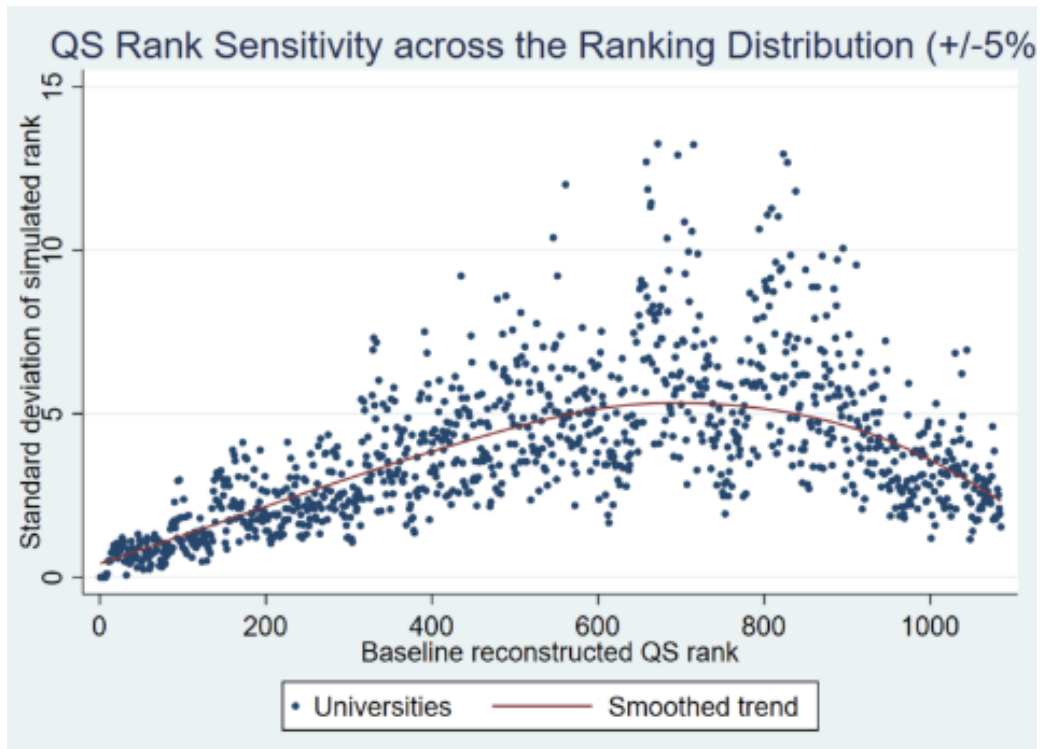


Figure 19: QS rank sensitivity across the ranking distribution under $\pm 5\%$ weight perturbation.

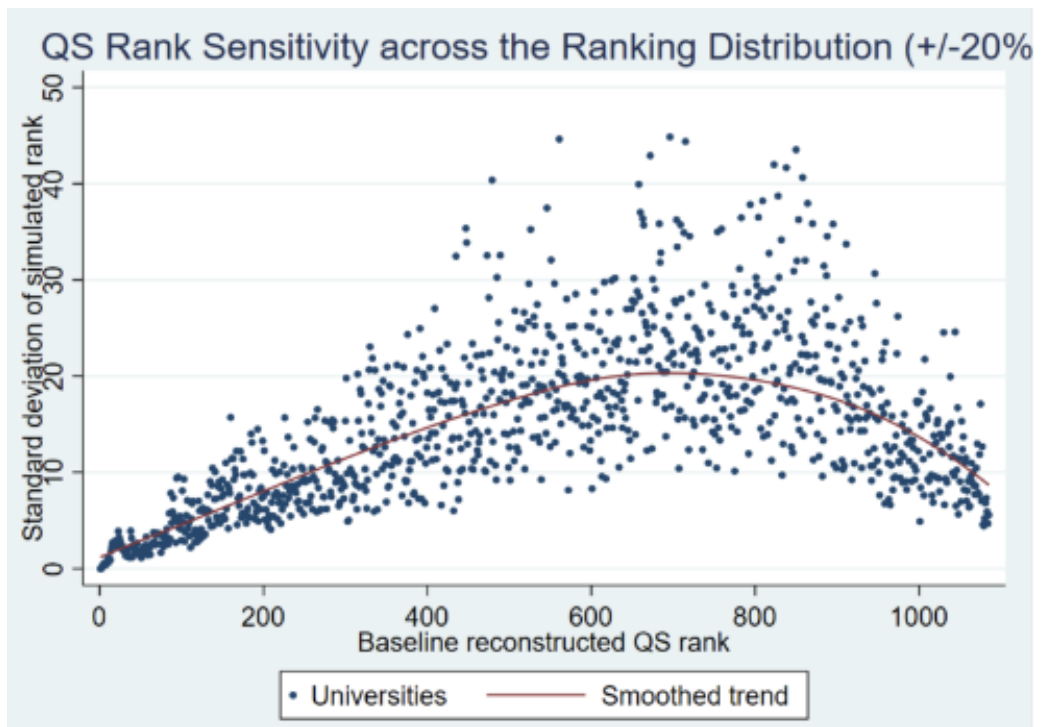


Figure 20: QS rank sensitivity across the ranking distribution under $\pm 20\%$ weight perturbation.

A.5 Practical Tool: Schema and User Access



Figure 21: Visual Schema of the practical tool.

A.5.1 User Access Management (UAM)

Access to the UPF Ranking Simulator is managed through Firebase Authentication. Users can access the tool by entering their authorised university email address and the corresponding access password. In the current implementation, access is restricted to users previously created by an administrator in the Firebase project.

To add a new authorised user, the administrator must follow these steps:

1. Access the Firebase Console of the application:
<https://console.firebase.google.com/project/upf-rankings/overview>

Only administrators with access to the Firebase project can enter this console using their UPF credentials.

2. Navigate to the **Authentication** section.
3. Open the **Users** tab and click on **Add user**.

4. Enter the **email address** of the user who should receive access and define an **access password**. For operational consistency, a common initial password may be used (**upf_EWE18**), although it should be shared only through internal channels and changed if required.

B Declaration on the Use of Generative Artificial Intelligence

In accordance with the regulations of Universitat Pompeu Fabra and the guidelines of the University Library/CRAI on the use of generative artificial intelligence (GenAI) tools in academic work, the authors of this Bachelor's Degree Final Thesis transparently declare the use made of such tools during the preparation of the work.

The authors take full responsibility for the truthfulness, integrity and correctness of all the material presented. The GenAI tools are not listed as authors or co-authors of the work, and the intellectual content, analysis, results and conclusions are the result of the authors' own work. All content generated or assisted by these tools has been reviewed, contrasted with specialised sources and critically validated by the authors.

The specific use made of these tools is detailed below:

Table 18: Details of the use of generative AI tools in the thesis.

Tool (model and version)	Purpose	Description of use
Claude Opus 4.8 (Anthropic)	Writing and style	Grammatical and spelling review, and improvement of the clarity and fluency of passages previously drafted by the authors.
Claude Sonnet 4.6 (Anthropic)	Programming support	Occasional help in debugging and commenting R/Stata code for the data analysis, always verified and adapted by the authors.

GenAI tools were in no case used to generate data, empirical results or entire sections of the work, nor to replace the authors' own analysis and interpretation. The full references for the tools used are included in the bibliography, in accordance with the citation style used in this thesis.

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